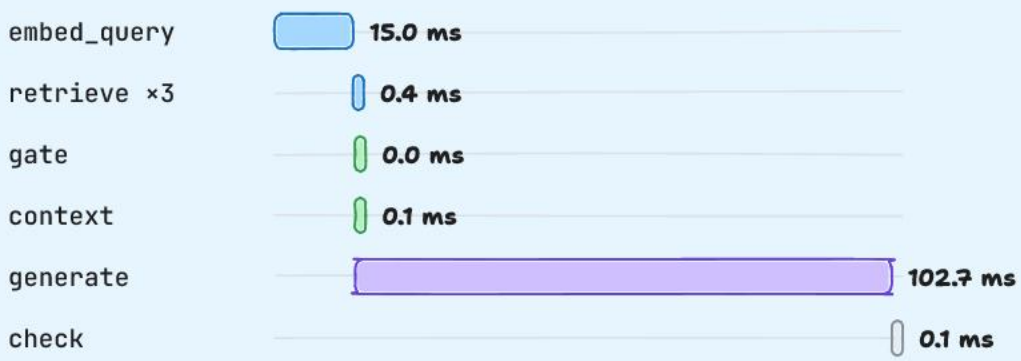
Trace every request

Day 13: where a request spends its time, and where it loses the answer.

ONE REQUEST, ONE TRACE

cold, one machine

"How soon must I report a lost laptop?"



ids, scores, decisions, times: no text

total 118.3 ms

the reader is 86% of a cold request; with vectors saved at index time, 1%

THE TRACE SAYS WHICH STEP LOST IT

sent `security-incident-reporting#how-to-report` ✓

cited `travel-policy#safety-while-travelling` ✗

retrieval found the answer; the reader picked another sentence

18 answerable questions show this pattern: sent, not cited

287 TRACES, NO TEXT IN ANY OF THEM

the question is a hash; documents, names and keys stay out

checked: no 6 words of a document, no 4 of the question

Time and money, per step

p50 and p95 per step; the cost of the requests that reach a model.

LATENCY: WHICH REQUEST?

one run, one machine

		p50	the biggest step
cold	nothing cached	119.56 ms	generate 86%
warm	sentence vectors saved once	12.10 ms	embed_query 96%
repeat	the answer cache	0.01 ms	the lookup
a model	Day 12's calls: ≈ 91 × warm	1.10 s	the model call

the biggest step changes with the request: measure each kind

p95s don't add up: the steps' 171.33 ms vs the requests' 167.05 ms (cold)

COST PER 1,000 REQUESTS THAT REACH A MODEL

`gpt-6-luna`	USD 0.10 / 0.50 per M tokens	USD 0.059
`gpt-6.1-sol`	USD 2 / 10 per M tokens	USD 1.187

the API-free reader: no token price, ≈ 12 ms of CPU per warm request

- the gate refuses 29 of 236 before any model call
- input is 76% of the cost; the sources are 58% of the input
- the provider's count ≈ 0.91 × our estimate (characters / 4)
- longest prompt ≈ 543 tokens: prompt caching starts at 1,024

prices checked 2026-11-12; tokens counted by the provider on 40 requests

Get the cache key right

Exact caches changed no answer; a missing part of the key did.

THE ANSWER CACHE'S KEY

groups | question | index fingerprint | pipeline version

+ re-check the user may still read every cited source

LEAVE ONE PART OUT

no groups — LEAK

maria (a manager) asks; alice asks it too

alice gets a managers-only answer

no fingerprint — STALE

the password policy: 14 → 16 characters

the cache still says 14

A SEMANTIC CACHE, DECIDED IN ADVANCE

serve a past answer if cosine ≥ 0.89: the lowest with no dev answer worse

51 new questions:



9 same outcome

0 better



1 worse

"If I give money to a charity, will Acme match it?"
nothing answers it; the pipeline refuses, the cache served an answer

value −2 at cost 2: not adopted; similar isn't the same

Plan for the model to fail

Timeouts, retries, a deadline and a breaker, tested on a fake server.

WHAT THE CLIENT DOES

real settings; the demo runs 20× faster

503, 500, 429	retry after a random wait (full jitter)
429 + Retry-After	wait as told, if it fits the deadline
401, no credit, not JSON	no retry: it won't pass
slow server	10 s per attempt, 3 attempts
trickling server	a 30 s deadline for the whole call
still failing	the fallback; the trace says why

a timeout is per read: Day 06's client waited 2.44 s with a 0.5 s timeout

A CIRCUIT BREAKER IN AN OUTAGE

8 requests



3 failed requests open it; after a cool-down, one good trial closes it

WHICH FALLBACK? DECIDED BY VALUE

the API-free reader instead of "unavailable", on 51 new questions:

20 correct · 7 wrong · 12 unanswerable ones answered

`correct +1 · refused 0 · wrong -2`

value −18: not adopted; say "unavailable" instead